CHARLES DARWIN UNIVERSITY AUSTRALIA

CDU IT Code Fair 2026 · Poster Competition
HIT401 Capstone · Group 13

CYBER SECURITY × GENERATIVE AI

AI is sharpening cyber attacks. We sharpened the defence.

We built an **AI security analyst** that investigates intrusion alerts and **writes its own detection rules** to catch attacks in network traffic, then honestly measured how well it does the job.

Automated Generation of Intrusion Detection Rules Using Generative AI in a Rule-Based IDS Deployment

13/14
attack types identified by the AI analyst
CIC-IDS2017 dataset

99.4%
of its rules' alerts were real attacks
stock rules: 6.1%

8/9
attack types found on a dataset it had not seen before
UNSW-NB15 dataset

0
false alarms from its rules on unseen normal traffic
held-out test day

Abstract

Intrusion detection only catches what its rules describe. In our open-source pipeline an AI model plays the **security analyst**: it investigates alerts, then writes and tests rules. We scored it on answer keys it never saw.

Background

AI is expected to make attacks more frequent and effective (NCSC, 2024), while defenders drown in alerts.

Our question: can the same AI do the analyst's job, and what does it need to know?

Objectives

- **Build** an AI analyst
- **Measure** every rule it writes
- **Test** what context it needs
- **Compare** it with human experts' rules

Methodology HOW IT WORKS

1 Traffic — Recorded attacks from two public datasets →
2 Suricata IDS — Open-source engine raises alerts →
3 AI analyst — Investigates each alert with 13 tools →
Threat → new detection rule — Tested by replay, then deployed
Harmless → silence the alert — Only if a **human approves**

How it became agentic

Think plans its next step → **Call a tool** picks 1 of 13 → **Engine checks** then runs it → **Read result** and repeats

↻ Repeats until a verdict (max 100 turns). Hunt rules must stay **quiet on normal traffic**.

Investigate	`query_eve_json` · `count_alerts` · `count_by` · `read_file` · `list_directory`
Hunt	`stack_values`
Research	`search_rules` · `lookup_vulnerability` · `suricata_keywords`
Validate	`run_suricata` · `test_rule`
Write	`write_file` · `append_to_file`

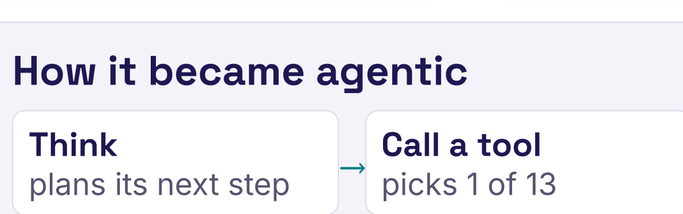
Tech stack

- **Suricata 8** IDS engine
- **ET Open** + 3 more open rulesets
- **Python** agentic triage engine
- **DeepSeek V4 Pro** & **Claude**, one adapter
- **FastAPI** web UI, streams each step live
- **Local NVD** vulnerability database

Results PRECISION

Is the alert a real attack?

CIC-IDS2017, noisiest capture days

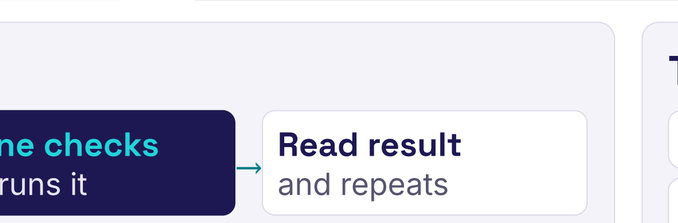


With full context it could tell **if an attack succeeded**: 52 times vs 9.

Results VS HUMANS

Gain over a strong human baseline

Unseen test day; baseline caught 29.5%



Hunting like **real detection engineers** worked best, with **zero** false alarms.

Analyst moments

Fixed its own mistake
First rule failed; it found why, fixed it.

Found what we missed
Spotted a botnet our report had missed.

Precise, not noisy
One rule: 1% → 28%, no false alarms.

Conclusions

Can AI replace the security analyst? **Our answer is split.**

Yes, for triage
Investigates alerts end to end and explains them.

Partly, for detection
Real gains on unseen data; risky changes need a human.

Honest limits

- Recorded traffic, not a live network
- Detects only, does not block
- Made confident mistakes

Scan to see the AI analyst at work
`triage.bigyan.work`

Team: Bigyan Siwakoti · Binay Siwakoti · Abisekh Kandel · Shashwat Acharya | **Supervisor:** Prof. Bharanidharan Shanmugam

Sharafaldin et al. (2018) CIC-IDS2017 · Moustafa & Slay (2015) UNSW-NB15 · NCSC (2024) The near-term impact of AI on the cyber threat.